File permissions in Linux

Project description

The research team needs its project files to match the organization's access rules. My task was to check the permissions in the projects directory, find anything that gave more access than allowed, and fix it from the Linux command line. This follows the principle of least privilege: each user should have only the access they need.

Check file and directory details

I moved into the projects directory and ran `ls -la`. The `-l` option lists permissions, and the `-a` option also shows hidden files, whose names start with a period.

```
cd projects
ls -la
```

```
researcher2@7d2059211ab9:~$ cd projects
researcher2@7d2059211ab9:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Oct 14 09:32 .
drwxr-xr-x 3 researcher2 research_team 4096 Oct 14 09:51 ..
-rw--w---- 1 researcher2 research_team   46 Oct 14 09:32 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Oct 14 09:32 drafts
-rw-rw-rw- 1 researcher2 research_team   46 Oct 14 09:32 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Oct 14 09:32 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Oct 14 09:32 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Oct 14 09:32 project_t.txt
researcher2@7d2059211ab9:~/projects$ cdmod
```

The directory contains five files, including one hidden file, and one subdirectory named drafts. Their permissions are:

Describe the permissions string

1st character: This character is either a d or hyphen (-) and indicates the file type. If it's a d, it's a directory. If it's a hyphen (-), it's a regular file.

● 2nd-4th characters: These characters indicate the read (r), write (w), and execute (x) permissions for the user. When one of these characters is a hyphen (-) instead, it

indicates that this permission is not granted to the user.
● 5th-7th characters: These characters indicate the read (r), write (w), and execute (x) permissions for the group. When one of these characters is a hyphen (-) instead, it indicates that this permission is not granted for the group.
● 8th-10th characters: These characters indicate the read (r), write (w), and execute (x) permissions for other. This owner type consists of all other users on the system apart from the user and the group. When one of these characters is a hyphen (-) instead, that indicates that this permission is not granted for other.

The current file permissions inside project:

User = read, write,
Group = read, write
Other = read, write

project_m.txt
User = read, write
Group = read
Other = none

project_r.txt
User= read, write
Group = read, write
Other = read
project_t.txt
User = read, write
Group = read, write
Other = read

project_x.txt
User = read, write
Group = write
Other = none

Change file permissions

The organization does not allow others to have write access to any files. Based on the permissions established in Step 3, identify which file needs to have its permissions modified.

The file project_k.txt is the only file that allows other to write, so we change it using this command

```
researcher2@7d2059211ab9:~/projects$ chmod o-w project_k.txt
researcher2@7d2059211ab9:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Oct 14 09:32 .
drwxr-xr-x 3 researcher2 research_team 4096 Oct 14 09:51 ..
-rw--w---- 1 researcher2 research_team   46 Oct 14 09:32 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Oct 14 09:32 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Oct 14 09:32 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Oct 14 09:32 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Oct 14 09:32 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Oct 14 09:32 project_t.txt
researcher2@7d2059211ab9:~/projects$
```

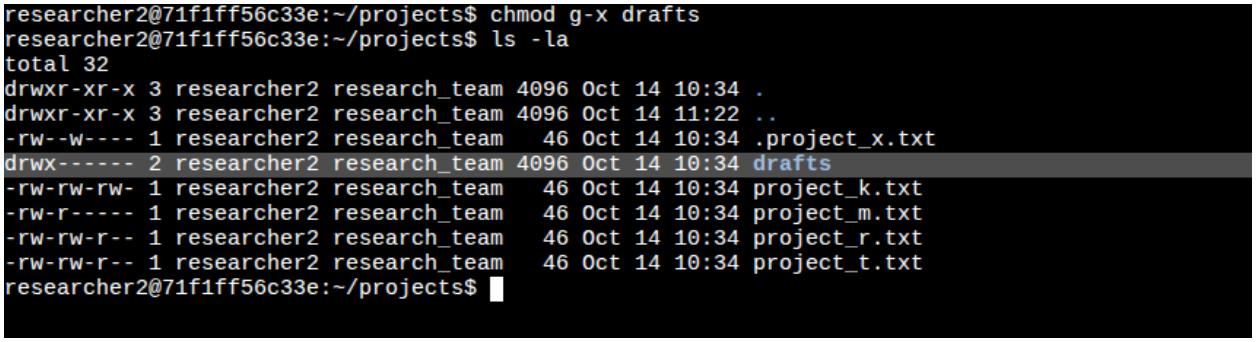
Change file permissions on a hidden file

The research team has archived **.project_x.txt**, which is why it's a hidden file. This file should not have write permissions for anyone, but the user and group should be able to read the file.
First we remove the write permission from user and group, then we add read to group

```
researcher2@7d2059211ab9:~/projects$ chmod u-w .project_x.txt
researcher2@7d2059211ab9:~/projects$ chmod g-w .project_x.txt
researcher2@7d2059211ab9:~/projects$ chmod g+r .project_x.txt
researcher2@7d2059211ab9:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Oct 14 09:32 .
drwxr-xr-x 3 researcher2 research_team 4096 Oct 14 09:51 ..
-r--r----- 1 researcher2 research_team   46 Oct 14 09:32 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Oct 14 09:32 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Oct 14 09:32 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Oct 14 09:32 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Oct 14 09:32 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Oct 14 09:32 project_t.txt
researcher2@7d2059211ab9:~/projects$
```

Change directory permissions

The files and directories in the projects directory belong to the **researcher2** user. Only **researcher2** should be allowed to access the **drafts** directory and its contents.
Only groups have execute permission, so by changing that, we will only allow users to read, write and execute, while group and others have no permissions.



Summary

I checked the permissions in the projects directory, including a hidden file, against the organization's rules. Three items gave more access than allowed. I used chmod to remove write access for other from project_k.txt, make .project_x.txt read-only for the user and group, and limit the drafts directory to its owner. Each change reduced access to only what the team needs, following the principle of least privilege.